Database System Engineering and Distributed Backend Development

WEEK-3

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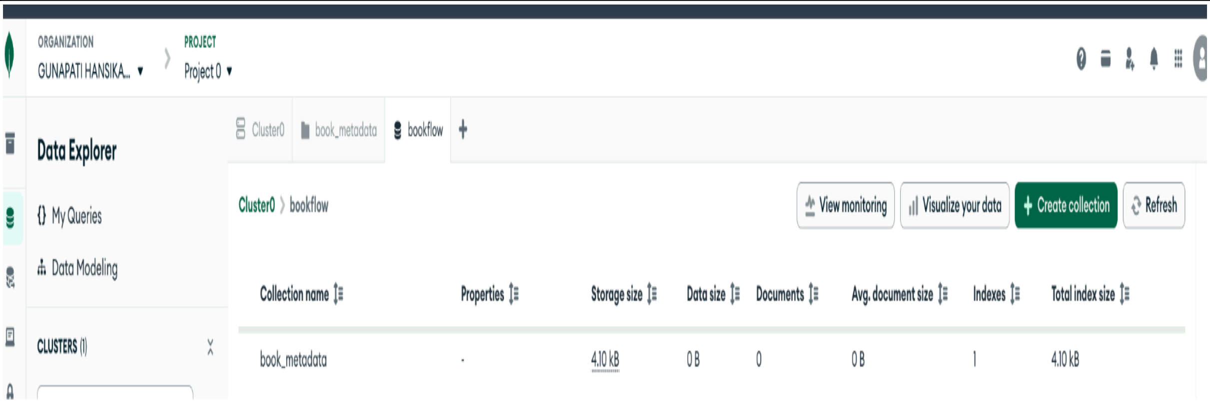
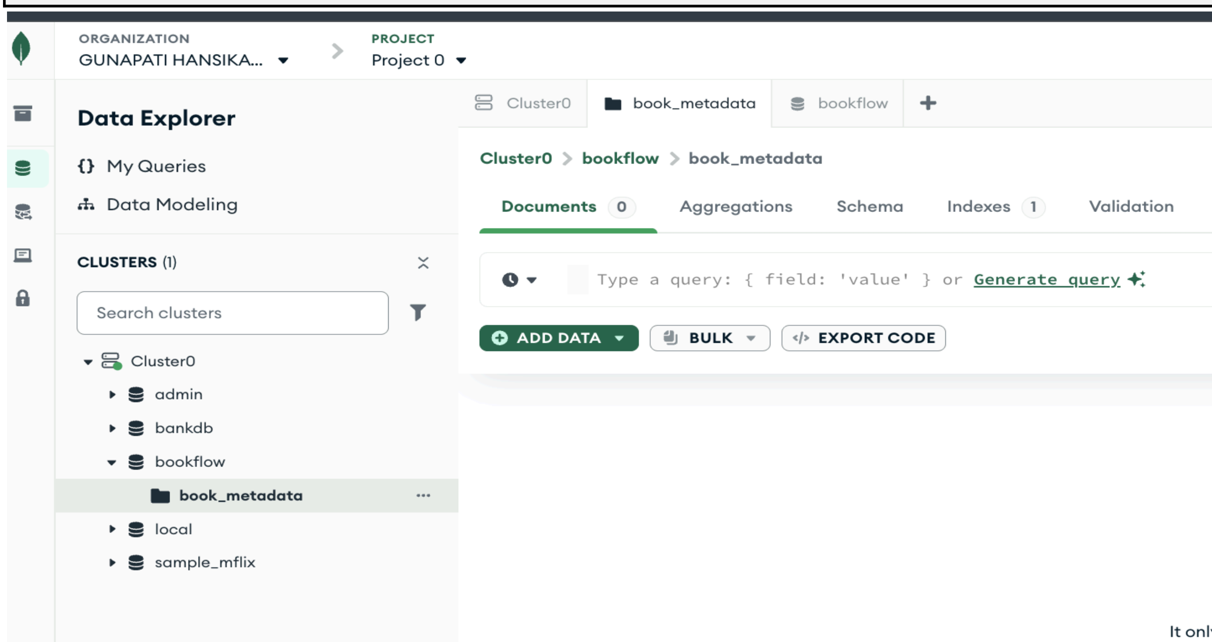
Step 1 — Switch to the BookFlow Database

```
use bookflow
```



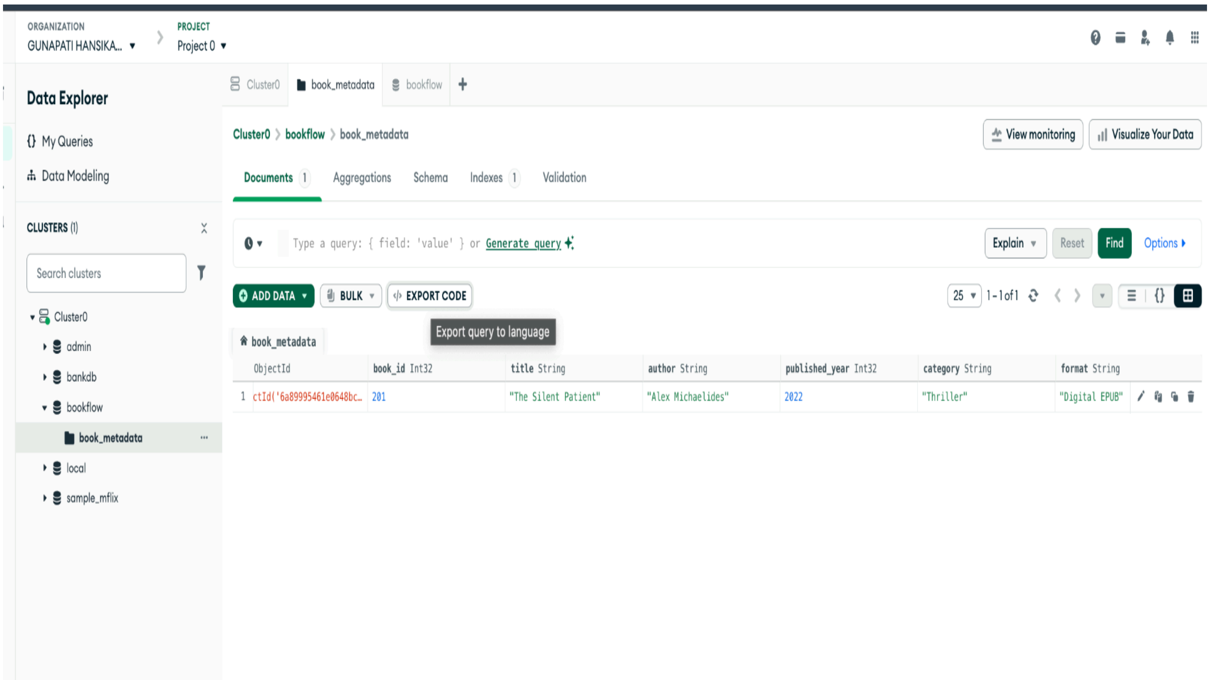
Step 2 — Create the book_metadata Collection

```
db.createCollection("book_metadata")
```



Step 3 — Insert Document 1 (Embedded + Referenced Pattern)

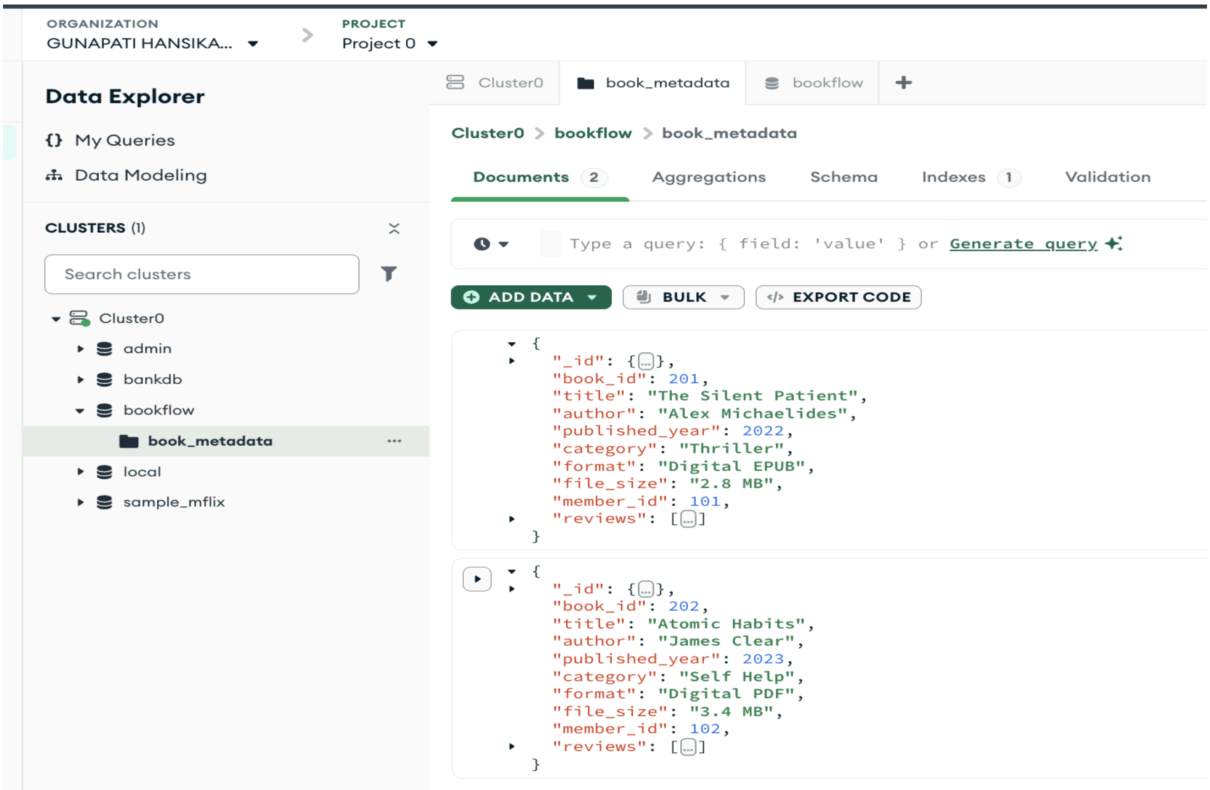
```
db.book_metadata.insertOne({
  book_id: 201,
  title: "The Silent Patient",
  author: "Alex Michaelides",
  published_year: 2022,
  category: "Thriller",
  format: "Digital EPUB",
  file_size: "2.8 MB",
  member_id: 101,
  reviews: [
    { member_id: 101, reviewer_name: "Hansika", rating: 5, comments: "Excellent
psychological thriller" },
    { member_id: 104, reviewer_name: "Aditya Rao", rating: 4, comments: "Very
gripping with an unexpected twist" },
    { member_id: 105, reviewer_name: "Sneha Kapoor", rating: 5, comments:
"Excellent narration and pacing" }
  ]
})
```



Step 4 — Insert Document 2 (Digital Format)

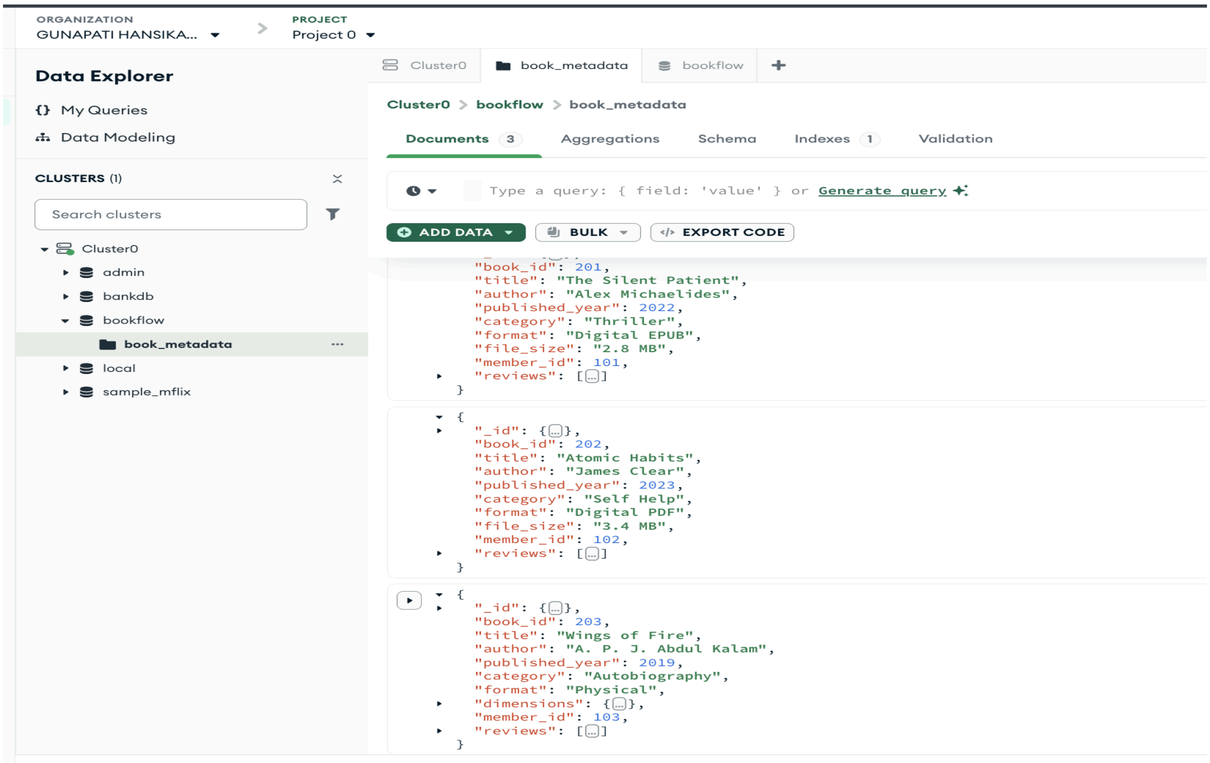
```
db.book_metadata.insertOne({
  book_id: 202,
  title: "Atomic Habits",
  author: "James Clear",
  published_year: 2023,
  category: "Self Help",
  format: "Digital PDF",
  file_size: "3.4 MB",
  member_id: 102,
  reviews: [
```

```
    { member_id: 102, reviewer_name: "Hasini", rating: 5, comments: "Excellent practical habit building guide" },
    { member_id: 106, reviewer_name: "Riya Sharma", rating: 4, comments: "Good but a little repetitive later on" },
    { member_id: 101, reviewer_name: "Hansika", rating: 5, comments: "Excellent life changing concepts" }
  ]
})
```



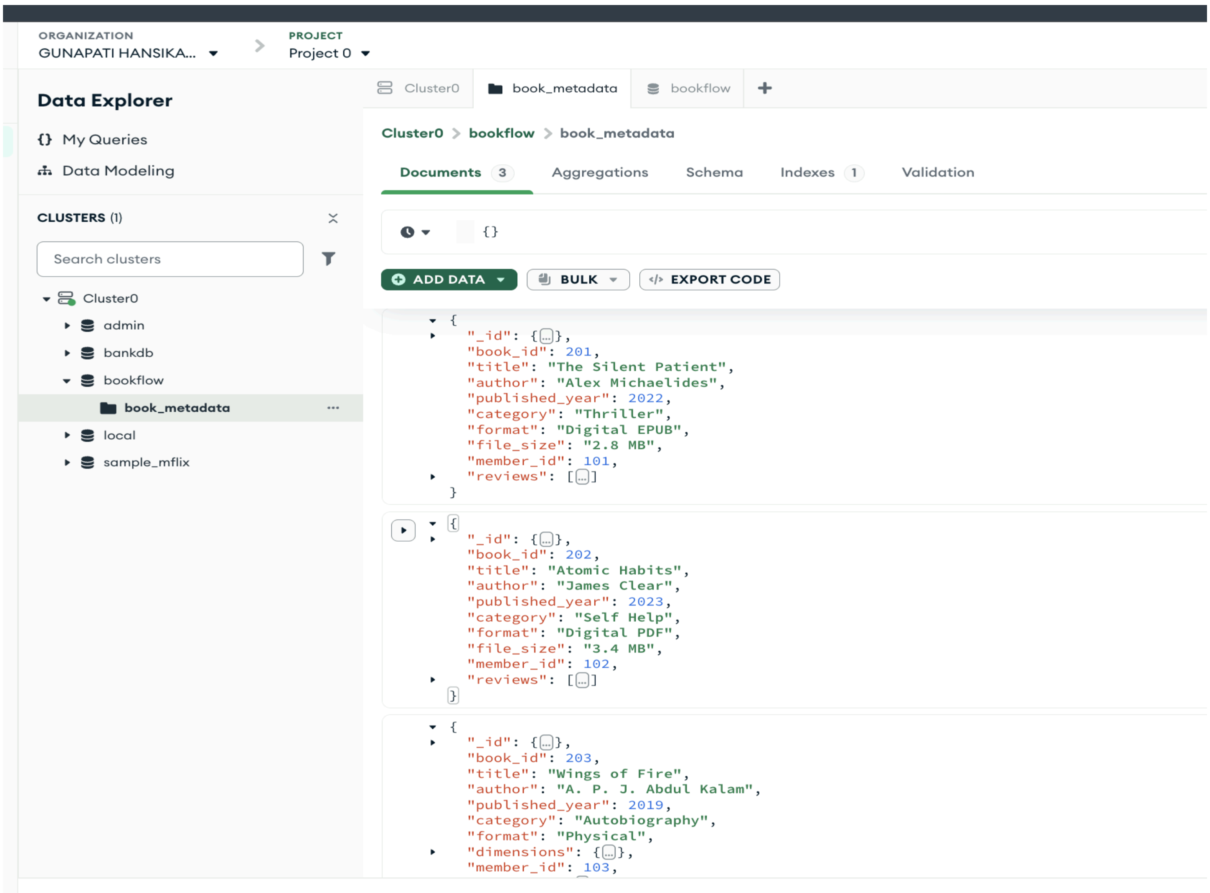
Step 5 — Insert Document 3 (Physical Format / Flexible Schema)

```
db.book_metadata.insertOne({
  book_id: 203,
  title: "Wings of Fire",
  author: "A. P. J. Abdul Kalam",
  published_year: 2019,
  category: "Autobiography",
  format: "Physical",
  dimensions: { height: "22 cm", width: "14 cm", weight: "350 g" },
  member_id: 103,
  reviews: [
    { member_id: 103, reviewer_name: "Bindhu", rating: 5, comments: "Excellent inspiring autobiography" },
    { member_id: 107, reviewer_name: "Kunal Mehta", rating: 4, comments: "Very motivational read for students" }
  ]
})
```



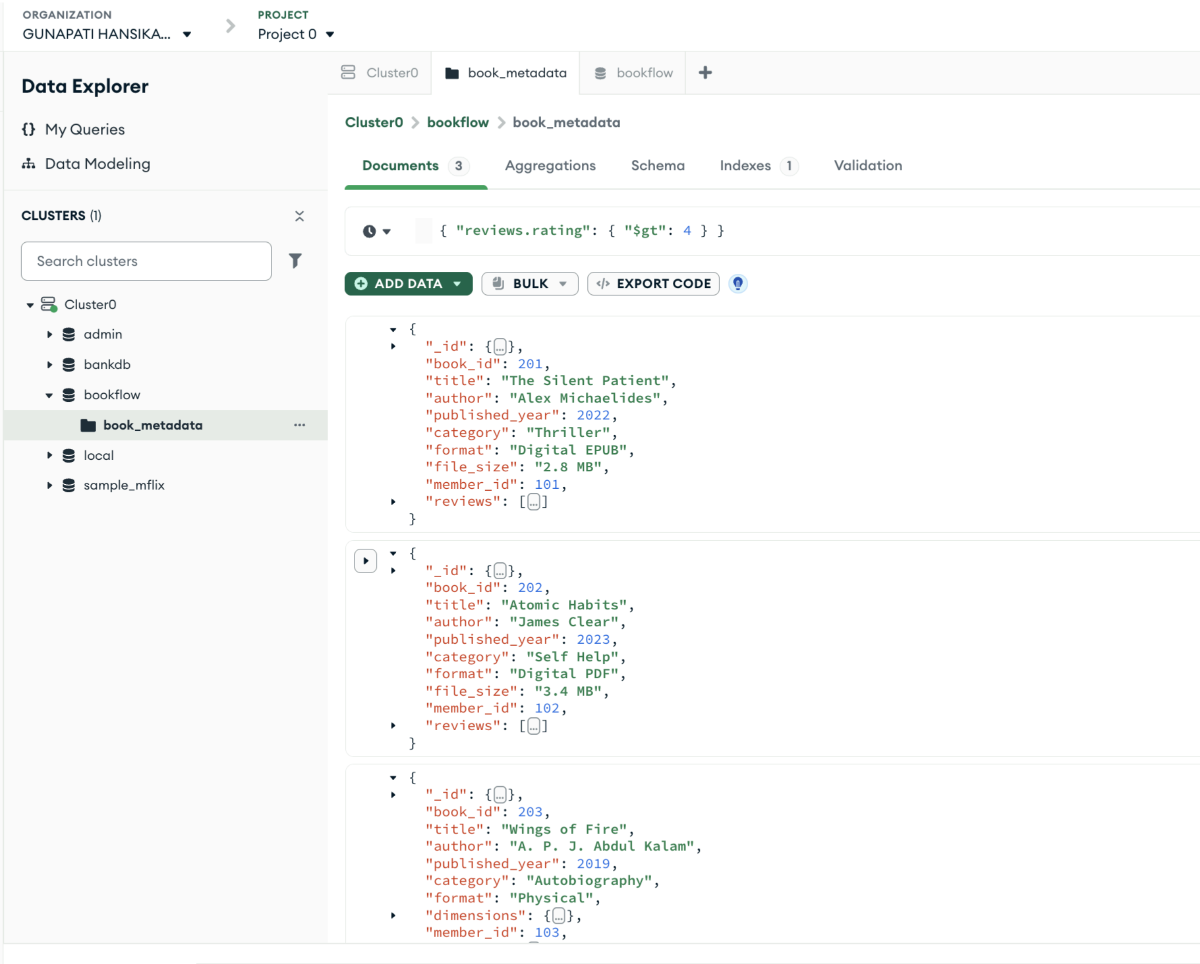
Step 6 — Verify Inserted Documents

```
db.book_metadata.find().pretty()
```



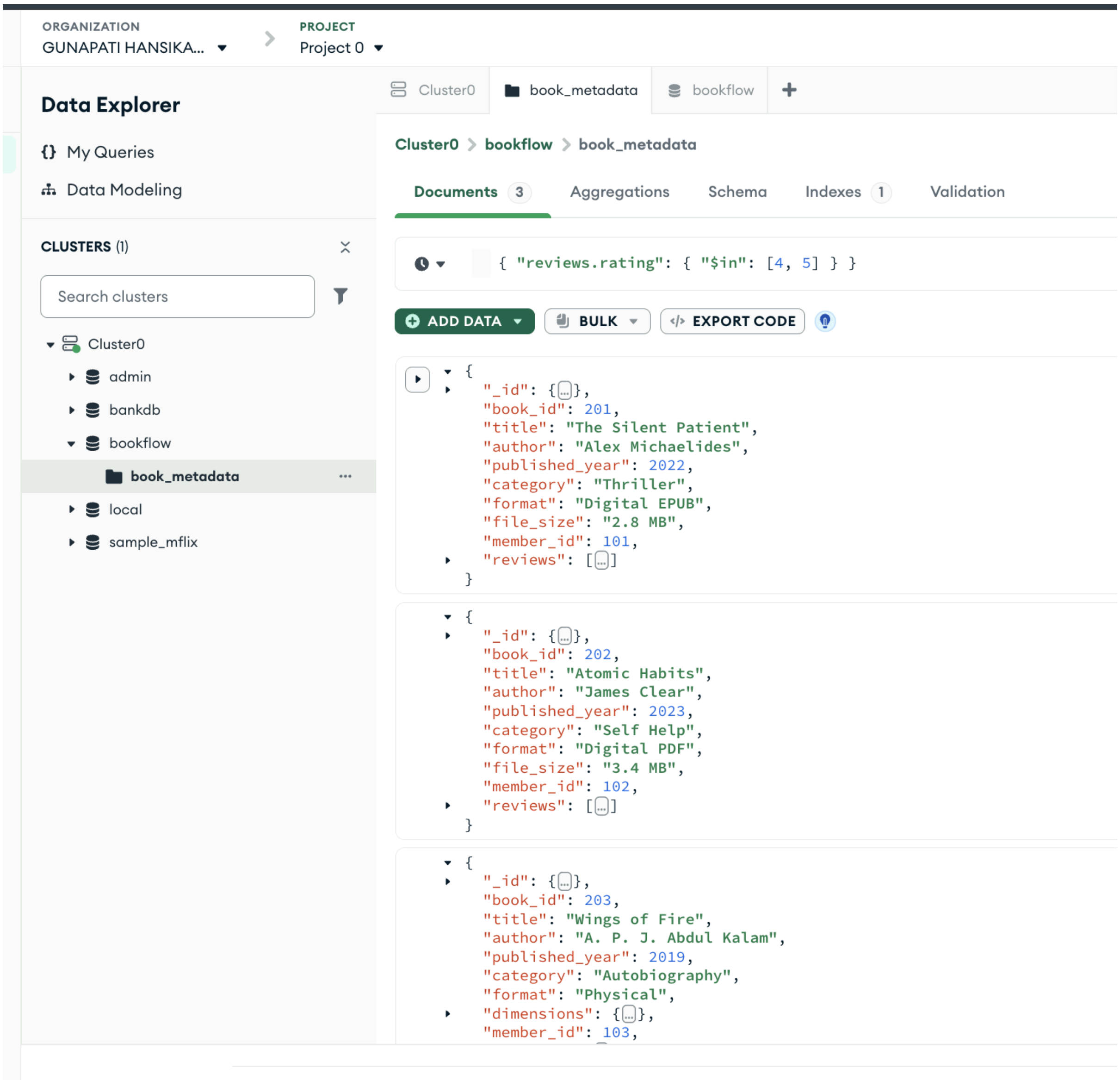
Step 7 — Comparison Query: $gt (Rating Greater Than 4)

```
db.book_metadata.find({
  "reviews.rating": { $gt: 4 }
})
```



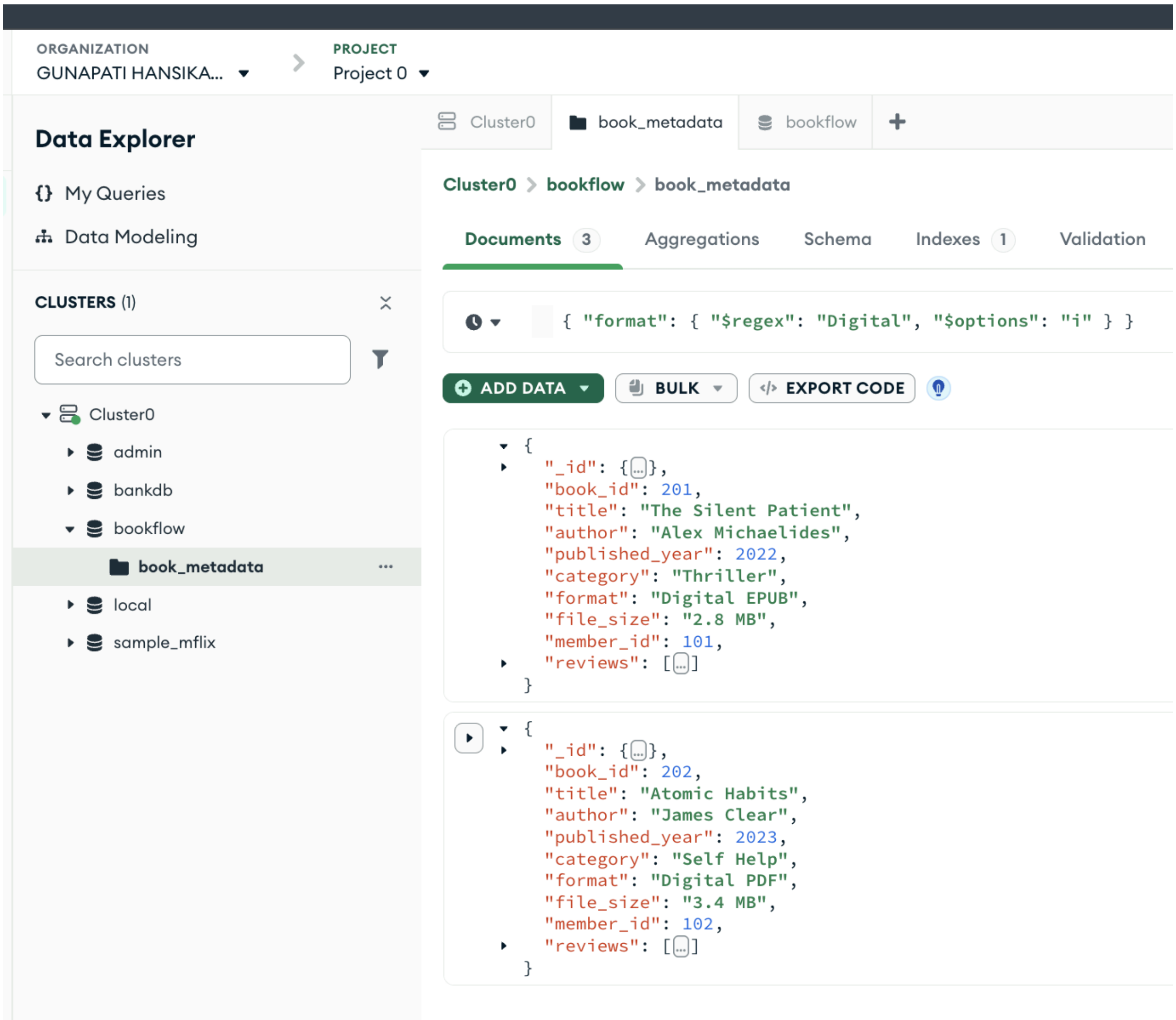
Step 8 — $in Query (Rating 4 or 5)

```
db.book_metadata.find({
  "reviews.rating": { $in: [4, 5] }
})
```



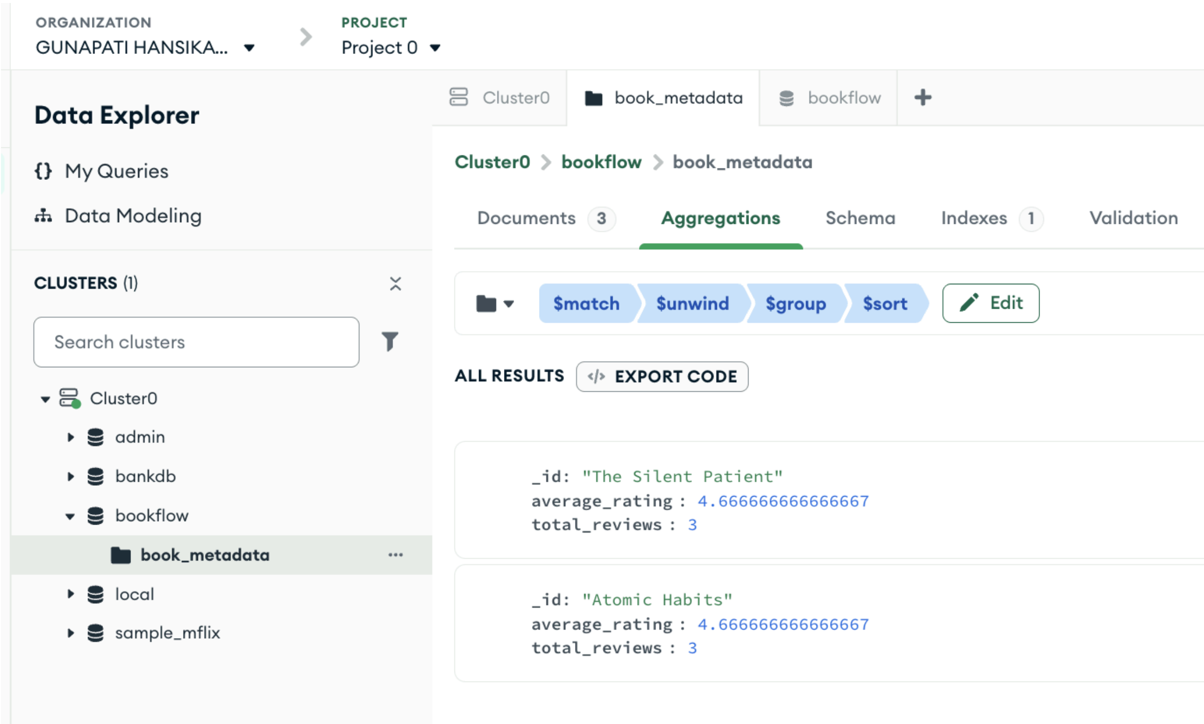
Step 9 — Regex Query (Digital Format)

```
db.book_metadata.find({
  format: { $regex: "Digital", $options: "i" }
})
```



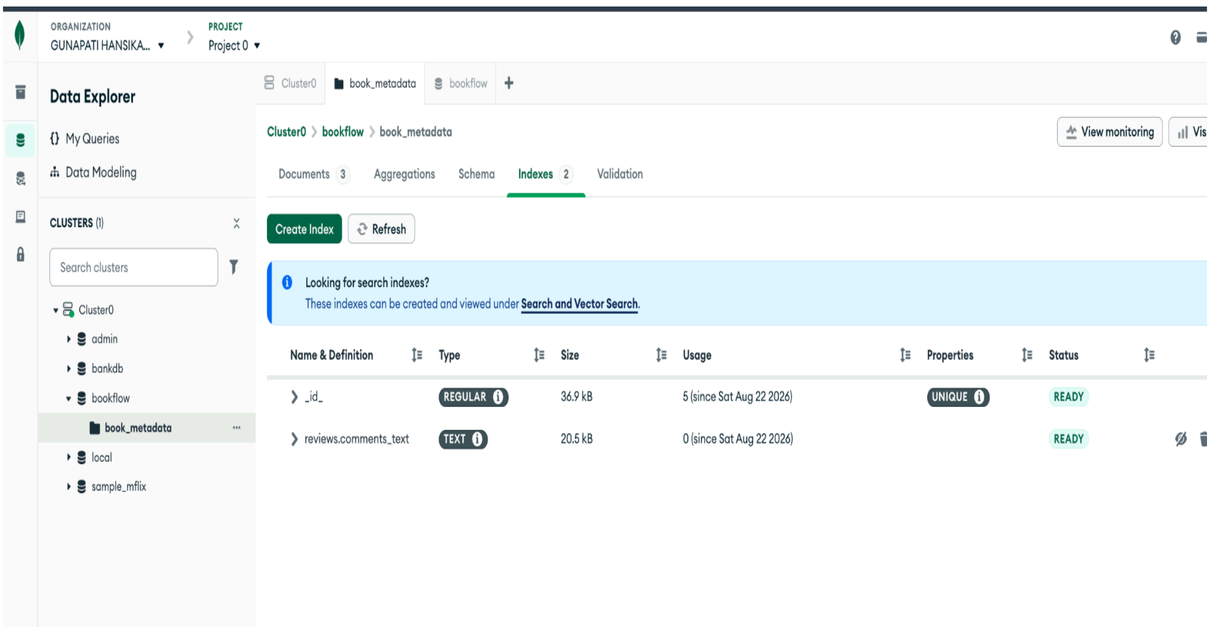
Step 10 — Aggregation Pipeline: Top Rated Books Report

```
db.book_metadata.aggregate([
  { $match: { published_year: { $gt: 2020 } } },
  { $unwind: "$reviews" },
  { $group: {
      _id: "$title",
      average_rating: { $avg: "$reviews.rating" },
      total_reviews: { $sum: 1 }
  } },
  { $sort: { average_rating: -1 } }
])
```



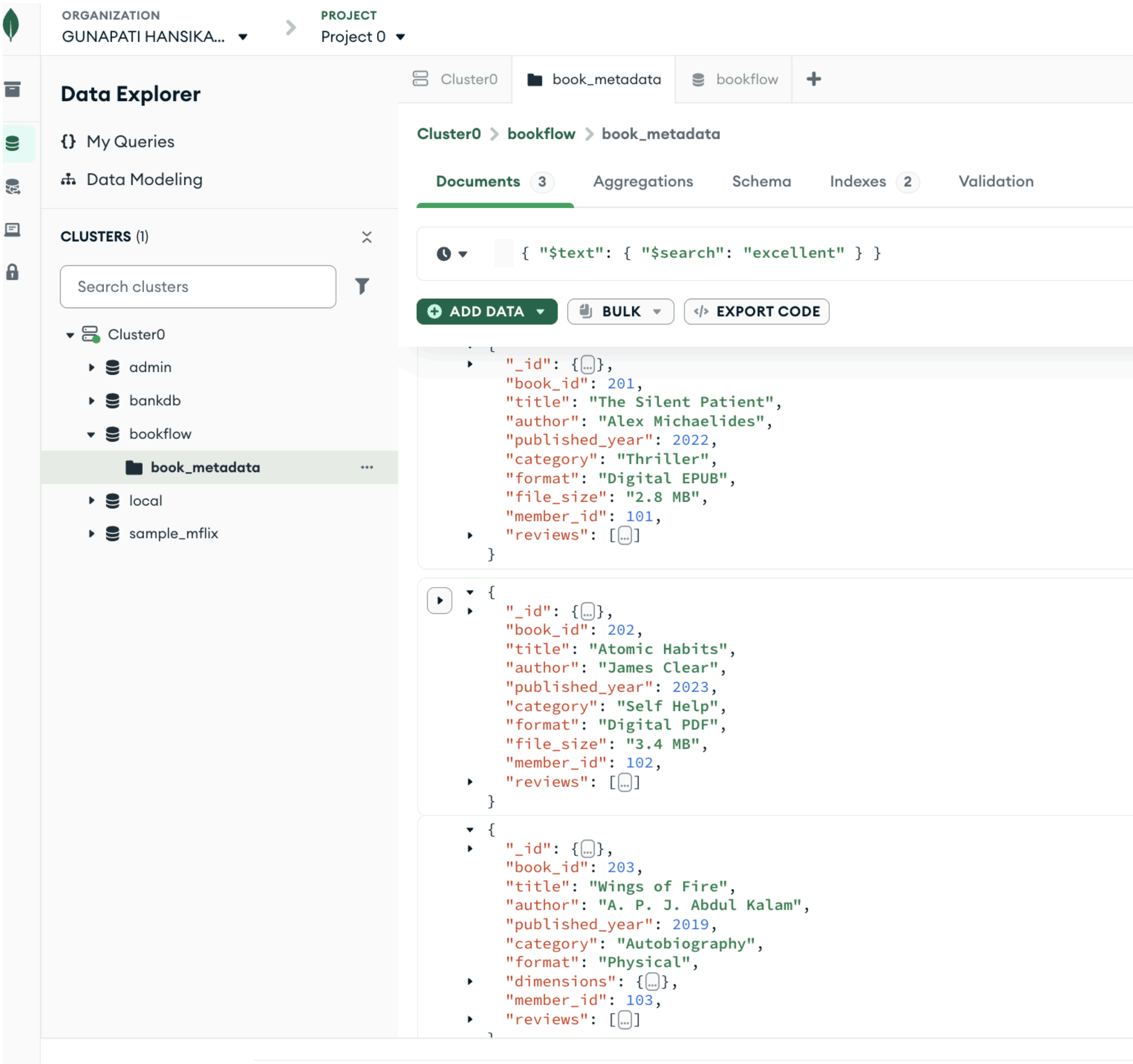
Step 11 — Create a Text Index on Review Comments

```
db.book_metadata.createIndex({
  "reviews.comments": "text"
})
```



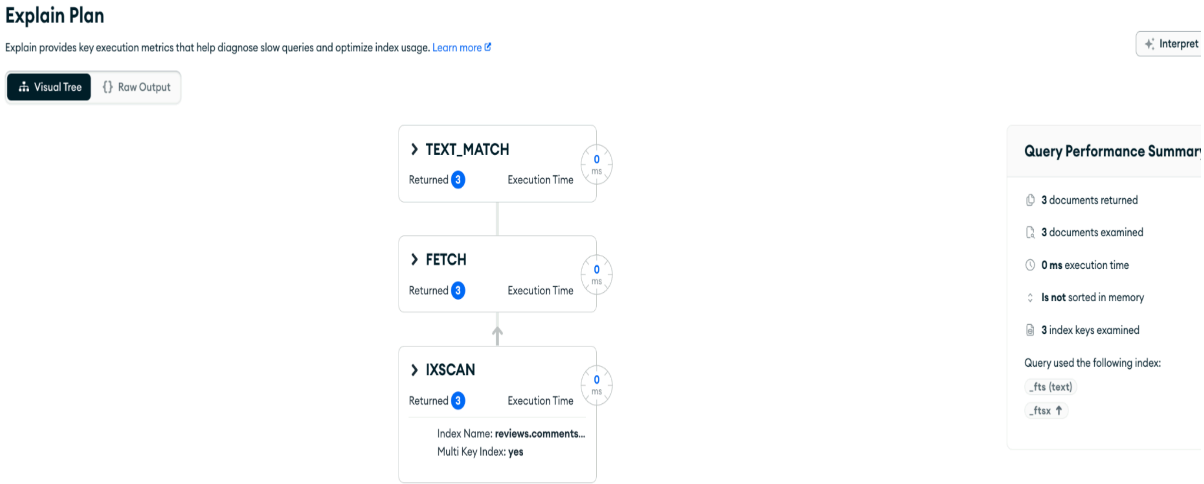
Step 12 — Text Search Query

```
db.book_metadata.find({
  $text: { $search: "excellent" }
})
```



Step 13 — explain() Query Performance Analysis

```
db.book_metadata.find({
  $text: { $search: "excellent" }
}).explain("executionStats")
```



Explain Plan

Explain provides key execution metrics that help diagnose slow queries and optimize index usage. Learn more

Visual Tree | {} Raw Output


```
{
  "explainVersion": "1",
  "queryPlanner": {
    "namespace": "bookflow.book_metadata",
    "parsedQuery": {
      "$text": {
        "$search": "excellent",
        "$language": "english",
        "$caseSensitive": false,
        "$diacriticSensitive": false
      }
    },
    "indexFilterSet": false,
    "queryHash": "CF6F4CEE",
    "planCacheShapeHash": "CF6F4CEE",
    "planCacheKey": "08852285",
    "optimizationTimeMillis": 0,
    "maxIndexedOrSolutionsReached": false,
    "maxIndexedAndSolutionsReached": false,
    "maxScansToExplodeReached": false,
    "prunedSimilarIndexes": false,
    "winningPlan": {
      "isCached": false,
      "stage": "TEXT_MATCH",
      "indexPrefix": {},
      "indexName": "reviews.comments_text",
      "parsedTextQuery": {
        "terms": ["excel"],
        "negatedTerms": [],
        "phrases": [],
        "negatedPhrases": []
      },
      "textIndexVersion": 3,
      "inputStage": {
        "stage": "FETCH",
        "inputStage": {
          "stage": "IXSCAN",
          "keyPattern": {
```